SYSTEM SPECIFICATION & ENGINEERING REFERENCE

NIRMAN AI

Detailed Technical Stack & Architecture

Comprehensive inventory of frameworks, libraries, ML pipelines, geospatial toolchains, database topologies, and deployment infrastructure.

Platform: NIRMAN AI — Decision Support System

Document Type: Technical Stack & Engineering Specification

Release / Date: October 2026 (Production POC Ready)

Target Region: Chennai Metropolitan Area, Tamil Nadu, India

Frontend Stack: React 19.2 + Vite 8.2 + Tailwind 4.3 + Three.js + Leaflet

Backend Stack: FastAPI 0.115 + Pydantic 2.10 + SQLAlchemy 2.0

Intelligence Layer: XGBoost 3.2 + SHAP 0.51 + Scikit-Learn (TF-IDF/SVD RAG)

Geospatial Layer: GeoPandas 1.1 + Shapely 2.1 + PostGIS 3.4

Deployment Target: Docker Compose / Render Cloud (Singapore)

Prepared by: NIRMAN AI Core Engineering Team

Status: Verified against Active Codebase & Lockfiles

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Architecture Overview

```
graph TD
    subgraph Client["■■ Client Layer"]
        REACT["React 19.2"]
        VITE["Vite 8.2"]
        TAILWIND["Tailwind CSS 4.3"]
        LEAFLET["Leaflet 1.9 + React-Leaflet 5.0"]
        RECHARTS["Recharts 3.10"]
        THREEJS["Three.js 0.185"]
        AXIOS["Axios 1.20"]
        JSPDF["jsPDF 4.2"]
        LUCIDE["Lucide React 1.42"]
        ROUTER["React Router DOM 7.18"]
    end

    subgraph Server["■■ Server Layer"]
        FASTAPI["FastAPI 0.115"]
        PYDANTIC["Pydantic 2.10"]
        SQLA["SQLAlchemy 2.0"]
        UVICORN["Uvicorn 0.34"]
    end

    subgraph Intelligence["■ Intelligence Layer"]
        SKLEARN["scikit-learn 1.9"]
        XGBOOST["XGBoost 3.2"]
        SHAP["SHAP 0.51"]
        PANDAS["Pandas 3.0"]
        NUMPY["NumPy 2.4"]
    end

    subgraph Spatial["■ Geospatial Layer"]
        GEOPANDAS["GeoPandas 1.1"]
        SHAPELY["Shapely 2.1"]
        POSTGIS["PostGIS 3.4"]
    end

    subgraph Data["■■ Data Layer"]
        PG["PostgreSQL 16"]
        SQLITE["SQLite"]
        PGVECTOR["pgvector"]
        MQTT["paho-mqtt 2.1"]
    end

    subgraph Infra["■■ Infrastructure"]
        DOCKER["Docker"]
        NGINX["nginx"]
        RENDER["Render"]
        MOSQUITTO["Mosquitto 2"]
    end

    Client -->|HTTP/JSON| Server
    Server --> Intelligence
    Server --> Spatial
    Server --> Data
    Data --> Infra
```

1. Frontend Stack

Core Framework

Technology	Version	Role	Why This Choice
React	19.2.8	UI framework	Component model, hooks, concurrent features, massive ecosystem
Vite	8.2.2	Build tool & dev server	Sub-second HMR, native ES modules, 10–100× faster than webpack
React Router DOM	7.18.3	Client-side routing	File-based routing patterns, nested layouts, data loading

Styling & Design

Technology	Version	Role	Why This Choice
Tailwind CSS	4.3.3	Utility-first CSS framework	Rapid prototyping, design consistency, zero unused CSS in production
Lucide React	1.42.0	Icon library	Tree-shakeable, consistent stroke icons, 1000+ icons

Data Visualisation

Technology	Version	Role	Why This Choice
Recharts	3.10.1	Charts & graphs	React-native, composable, built on D3.js primitives
Leaflet	1.9.4	Interactive maps	Lightweight (42 KB), OpenStreetMap, plugin ecosystem
React-Leaflet	5.0.0	React bindings for Leaflet	Declarative map components, hooks API
Three.js	0.185.1	3D visualisation	WebGL-powered 3D city visualisations

Utilities

Technology	Version	Role	Why This Choice
Axios	1.20.0	HTTP client	Interceptors, request/response transforms, automatic JSON
jsPDF	4.2.1	Client-side PDF	Browser-native PDF generation without server round-trip
jsPDF-AutoTable	5.0.8	PDF tables	Structured table rendering in client PDFs

Testing (Frontend)

Technology	Version	Role
Vitest	5.0.0	Test runner (Vite-native, Jest-compatible API)
@testing-library/react	16.3.3	Component testing (user-centric queries)
@testing-library/jest-dom	7.0.1	DOM assertion matchers
@testing-library/user-event	14.6.7	User interaction simulation
jsdom	29.1.1	Browser environment for Node.js
@vitest/coverage-v8	5.0.0	Code coverage via V8

Build Tooling

Technology	Version	Role
@vitejs/plugin-react	6.1.0	React Fast Refresh, JSX transform
@tailwindcss/vite	4.3.3	Tailwind CSS integration for Vite
oxlint	1.79.0	Linter (Rust-based, 50–100× faster than ESLint)

Frontend File Structure

```
frontend/src/
■■■ api/            # Axios client with base URL config and interceptors
■   ■■■ client.js
■■■ assets/         # Static assets (images, fonts)
■■■ components/     # Reusable UI components
■   ■■■ journey/    # Multi-step workflow components
■   ■■■ CityScape3D.jsx   # Three.js 3D city visualisation
■   ■■■ uikit.jsx         # Shared design system components
■■■ context/        # React Context providers
■   ■■■ ProjectContext.jsx
■■■ data/           # Client-side reference data
```

```
■■■ pages/          # 24 feature pages
■   ■■■ Dashboard.jsx         # Portfolio rollup
■   ■■■ SiteStudio.jsx        # Deep-dive site assessment (70 KB)
■   ■■■ DecisionStudio.jsx    # Integrated decision workspace (41 KB)
■   ■■■ Demand.jsx            # Population & demand analysis (34 KB)
■   ■■■ DeliveryStudio.jsx    # Construction planning (32 KB)
■   ■■■ Copilot.jsx           # Natural-language AI assistant (30 KB)
■   ■■■ Explorer.jsx          # Infrastructure inventory (28 KB)
■   ■■■ Recommendation.jsx    # Site ranking engine (26 KB)
■   ■■■ Priority.jsx          # Project prioritisation (24 KB)
■   ■■■ WhatIf.jsx            # Scenario simulator (21 KB)
■   ■■■ Dashboard.jsx         # Portfolio overview (22 KB)
■   ■■■ OperationsHub.jsx     # Operations management (18 KB)
■   ■■■ Dpr.jsx               # Detailed Project Report (15 KB)
■   ■■■ Cost.jsx              # Cost estimation (14 KB)
■   ■■■ MapPage.jsx           # Infrastructure map (12 KB)
■   ■■■ Iot.jsx               # IoT monitoring (11 KB)
■   ■■■ Schemes.jsx           # Government schemes (11 KB)
■   ■■■ Knowledge.jsx         # Knowledge base (10 KB)
■   ■■■ Landing.jsx           # Landing page (9 KB)
■   ■■■ Risk.jsx              # Risk classification (9 KB)
■   ■■■ Settings.jsx          # App settings (8 KB)
■   ■■■ Reports.jsx           # Reporting (8 KB)
■   ■■■ Plan.jsx              # Project planning (7 KB)
■   ■■■ DataLineage.jsx       # Data provenance (7 KB)
■   ■■■ Login.jsx             # Authentication (6 KB)
■■■ test/           # 23 component tests
■■■ index.css       # Global styles (20 KB)
■■■ theme.js        # Design tokens
■■■ main.jsx        # App entry point
■■■ App.jsx         # Root component with routing
```

2. Backend Stack

Web Framework

Technology	Version	Role	Why This Choice
FastAPI	0.115.6	REST API framework	Async-native, auto-generated OpenAPI docs, Pydantic integration, type safety
Uvicorn	0.34.0	ASGI server	High-performance, HTTP/2, WebSocket support
Pydantic	2.10.4	Data validation	Type-safe request/response contracts, serialisation, settings management
Pydantic Settings	2.7.0	Configuration	Environment variable parsing with type coercion and defaults
httpx	0.28.1	Async HTTP client	For outbound calls to LLM providers

Middleware Stack

```
Request → GZip Compression → CORS → ETag → Timing → Router → Response
```

Middleware	Purpose
GZipMiddleware	Compress responses > 500 bytes
CORSMiddleware	Cross-origin access control (configurable origins)
ETagMiddleware	Cache validation via content hashing
TimingMiddleware	Request duration tracking for observability

ORM & Database

Technology	Version	Role	Why This Choice
SQLAlchemy	2.0.36	ORM	Mature, supports both sync/async, dialect abstraction (SQLite ↔ PostgreSQL)
Alembic	1.14.0	Migrations	Schema versioning, auto-generation from models
psycopg	3.2.3	PostgreSQL driver	Native async, binary protocol, modern Python
python-dotenv	1.0.1	Environment loading	`.env` file parsing

Backend File Structure

```
backend/app/
■■■ api/v1/              # REST routers (all endpoints)
■■■ core/                # Configuration, database, constants, migrations, observability
■   ■■■ config.py        # Pydantic Settings (env vars)
■   ■■■ database.py      # SQLAlchemy engine, session, health check
■   ■■■ constants.py     # Disclaimer text, data status enums
■   ■■■ migrations.py    # PostGIS/pgvector column promotion
■   ■■■ observability.py # ETag, Timing middleware, metrics
■■■ engines/             # All scoring, ranking, and costing logic
■   ■■■ suitability.py   # 9-factor site scoring (16 KB)
■   ■■■ ml.py            # XGBoost + SHAP pipeline (16 KB)
■   ■■■ ml_training.py   # Model training utilities (8 KB)
■   ■■■ cost.py          # Planning-level cost estimation (8 KB)
■   ■■■ priority.py      # Multi-factor project ranking (7 KB)
■   ■■■ risk.py          # Flood/terrain/access classification (6 KB)
■   ■■■ schemes.py       # Government scheme matching (6 KB)
■   ■■■ whatif.py         # Scenario simulation (5 KB)
■   ■■■ explain.py        # Factor attribution (4 KB)
■   ■■■ demand.py         # Population growth & demand (4 KB)
■   ■■■ weights.py        # Weight validation (must total 100) (3 KB)
■   ■■■ normalize.py      # Score normalisation modes (3 KB)
```

```
■■■ models/              # SQLAlchemy ORM models
■   ■■■ planning.py      # Sites, projects, schemes, priorities (8 KB)
■   ■■■ spatial.py       # GIS layers, coordinates, boundaries (9 KB)
■   ■■■ documents.py     # RAG documents, chunks, embeddings (4 KB)
■   ■■■ iot.py           # Devices, readings, thresholds (3 KB)
■   ■■■ registry.py      # Data sources, lineage (2 KB)
■   ■■■ users.py         # User accounts (1 KB)
■■■ providers/
■   ■■■ ai/              # MockAIProvider, LocalLLMProvider, HostedLLMProvider
■   ■■■ data/            # DemoDataProvider + declared GCC/CMDA/OSM/Bhuvan/IMD/Census
■   ■■■ recommendation/  # MCDA (default) and ML+SHAP scoring strategies
■■■ repositories/        # The ONLY place SQL happens
■■■ schemas/             # Pydantic request/response contracts
■■■ seed/                # CSV/JSON seed data loader
■■■ services/            # Orchestration layer
■   ■■■ copilot_service.py     # Intent detection → engine → explanation (15 KB)
■   ■■■ dpr_service.py         # Detailed Project Report generator (21 KB)
■   ■■■ rag_service.py         # TF-IDF + SVD retrieval pipeline (17 KB)
■   ■■■ gis_service.py         # GeoPandas spatial analysis (12 KB)
■   ■■■ grievance_service.py   # Public grievance handling (11 KB)
■   ■■■ project_service.py     # Project lifecycle management (10 KB)
■   ■■■ site_service.py        # Site assessment orchestration (9 KB)
■   ■■■ mqtt_service.py        # MQTT subscriber + ingestion (7 KB)
■   ■■■ iot_service.py         # Device management + simulation (6 KB)
■   ■■■ persistence_service.py # Score/audit persistence (5 KB)
■   ■■■ supabase_service.py    # Cloud storage integration (4 KB)
■■■ utils/               # Shared utilities
```

3. Data Processing & ML Stack

Data Processing

Technology	Version	Role	Why This Choice
Pandas	3.0.5	Tabular data processing	Industry standard, DataFrame API, CSV/JSON I/O
NumPy	2.4.6	Numerical computation	Array operations, linear algebra, statistical functions

Machine Learning

Technology	Version	Role	Why This Choice
scikit-learn	1.9.0	ML utilities	Preprocessing, cross-validation, TF-IDF vectorisation, truncated SVD

Technology	Version	Role	Why This Choice
XGBoost	3.2.0	Gradient boosted trees	High accuracy, handles mixed feature types, fast inference
SHAP	0.51.0	Model explainability	Exact additive attributions via TreeExplainer, regulatory transparency

ML Pipeline Detail

```
graph LR
    subgraph Training
        DATA["40 Labelled Sites"] --> FEAT["15 Features Extracted"]
        FEAT --> XGB["XGBRegressor (depth=2, n=220)"]
        XGB --> CV["5-Fold Cross Validation"]
        CV --> METRICS["R²≈0.79, MAE≈1.0"]
    end

    subgraph Inference
        SITE["New Site"] --> SCORE["XGBoost Prediction"]
        SCORE --> SHAP_EXP["SHAP TreeExplainer"]
        SHAP_EXP --> ATTR["Additive Attributions"]
        ATTR --> VERIFY["base + Σcontributions = prediction"]
    end

    subgraph Fallback
        FAIL["Model Failure"] --> MCDA["Deterministic MCDA"]
        MCDA --> NOTE["Response includes degradation note"]
    end
```

RAG Pipeline (Knowledge Base)

Component	Technology	Detail
Vectorisation	scikit-learn TfidfVectorizer	Deterministic, no API key needed
Dimensionality Reduction	scikit-learn TruncatedSVD	Compress TF-IDF to dense vectors
Similarity	Cosine similarity	Top-k passage retrieval
Storage (PostgreSQL)	pgvector	`vector(N)` column with HNSW index
Storage (SQLite)	JSON column	Similarity computed in Python
Corpus	Platform methodology, scheme reference, data registry	Only what the platform can vouch for

4. Geospatial Stack

Technology	Version	Role	Why This Choice
GeoPandas	1.1.4	Geospatial DataFrames	Spatial joins, projections, buffer analysis
Shapely	2.1.2	Geometric operations	Point, polygon, distance, area computations
PostGIS	3.4	Spatial database extension	GiST indexes, geometry columns, spatial queries

Projection Strategy

Layer	CRS	Reason
Storage	WGS 84 (EPSG:4326)	Portable lat/lng — works on SQLite
Analysis	UTM 44N (EPSG:32644)	Metric distances and areas — "5 km" is genuinely 5 km
Display	WGS 84 (EPSG:4326)	Leaflet/OpenStreetMap standard

Spatial Analyses Performed

Analysis	Method	Output
Nearest neighbours	Projected distance matrix	Site separation in kilometres
Catchment overlap	5 km buffer intersection	Sites competing for same population
Coverage gaps	Set difference against site buffers	Underserved localities (e.g., Nemmeli at 10.6 km)
Convex hull	Shapely convex_hull	Study area extent and area

5. Database Stack

Dual-Database Strategy

```
graph LR
    subgraph Development
        APP_DEV["FastAPI"] --> SQLITE["SQLite"]
        SQLITE --> NOTE_DEV["Zero config, git clone to working demo"]
    end

    subgraph Production
        APP_PROD["FastAPI"] --> PG["PostgreSQL 16"]
        PG --> POSTGIS_EXT["PostGIS 3.4"]
        PG --> PGVEC["pgvector"]
        POSTGIS_EXT --> GEOM["geometry(Point, 4326) + GiST indexes"]
        PGVEC --> VEC["vector(N) + HNSW index"]
    end
```

Database Models

Model File	Tables	Purpose
`planning.py`	Sites, Projects, Schemes, Priorities, Scores	Core planning entities
`spatial.py`	GIS Layers, Coordinates, Boundaries, Catchments	Geospatial features
`documents.py`	Sources, Documents, DocumentChunks, Embeddings	RAG knowledge base
`iot.py`	Devices, Readings, Thresholds, Alerts	IoT sensor data
`registry.py`	DataSources, Lineage, Providers	Data governance
`users.py`	Users, Sessions	Authentication

Migration Strategy

Migration	File	Purpose
001	Schema creation	`Base.metadata.create_all()` at startup
002	`002_spatial_and_indexes.sql`	Promote lat/lng to `geometry(Point, 4326)`, add GiST indexes
003	`003_vector_columns.sql`	Promote JSON embeddings to `vector(N)`, add HNSW index

> **[NOTE]** Migrations 002 and 003 run at application startup (not Docker init) because they need the tables to exist first. On SQLite they are skipped gracefully.

6. IoT Stack

Technology	Version	Role	Why This Choice
ESP32	—	Microcontroller	Wi-Fi + BLE, low power, widely available, Arduino/MicroPython support
paho-mqtt	2.1.0	Python MQTT client	Standard MQTT library, QoS levels, TLS support
Eclipse Mosquitto	2	MQTT broker	Lightweight, Docker image, production-ready

Message Contract

```
Topic: nirman/sensors/{device_id}/{sensor_type}

Payload:
{
  "device_id": "ESP32-WL-001",
  "sensor_type": "water_level",
  "value": 2.4,
  "unit": "meters",
  "latitude": 12.9352,
  "longitude": 80.2190,
  "timestamp": "2026-10-12T14:00:00Z"
}
```

Ingestion Path

Source	Endpoint	`is_demo_data`
Real ESP32 via MQTT	`nirman/sensors/#` subscriber	`false`
HTTP ingest	`POST /api/v1/iot/ingest`	`false`
Simulator	`POST /api/v1/iot/simulate`	`true` (always)

7. PDF Generation Stack

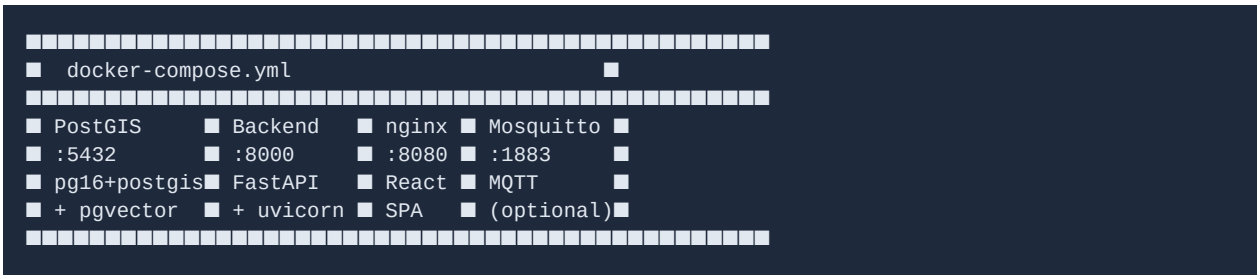
Layer	Technology	Version	Use Case
Server-side	ReportLab	4.2.5	DPR PDF generation with structured layouts, tables, and disclaimers
Client-side	jsPDF	4.2.1	Browser-native PDF export for reports and summaries
Client-side	jsPDF-AutoTable	5.0.8	Structured table rendering in client PDFs

8. Infrastructure & Deployment

Containerisation

Technology	Purpose	Configuration
Docker	Container runtime	Multi-stage builds for frontend and backend
Docker Compose	Multi-service orchestration	4 services: db, backend, frontend, mqtt (optional)
nginx	Reverse proxy	Frontend serving + `/api` proxying to backend

Docker Compose Services



Render Deployment (Cloud)

Setting	Value
Runtime	Docker (all-in-one Dockerfile)
Plan	Free tier

Setting	Value
Region	Singapore
Health check	`/health`
Auto-deploy	On git push
Database	Ephemeral SQLite (rebuilt from seed on each start)
Single origin	Frontend + API on same URL

Environment Variables

Variable	Default	Purpose
`DATABASE_URL`	SQLite file	PostgreSQL connection string for production
`CORS_ORIGINS`	`localhost:5173`	Comma-separated allowed origins
`DEMO_MODE`	`true`	Enable seeded demo data and simulated IoT
`AUTO_SEED`	`true`	Seed database on startup if empty
`AI_PROVIDER`	`mock`	`mock` / `local` / `hosted`
`LLM_BASE_URL`	unset	OpenAI-compatible LLM endpoint
`LLM_MODEL`	unset	Model name for LLM provider
`LLM_API_KEY`	unset	API key (environment only, never in source)
`MQTT_BROKER_URL`	unset	MQTT broker for IoT (optional)
`JWT_SECRET`	placeholder	JWT signing key (auto-generated on Render)

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9. Testing Stack

Backend (Python)

Tool	Purpose	Count
pytest	Test runner and framework	106+ tests across 7 test files

Tool	Purpose	Count
conftest.py	Shared fixtures (test database, client)	Test DB with in-memory SQLite

Frontend (JavaScript)

Tool	Purpose	Count
Vitest	Vite-native test runner	23 component tests
React Testing Library	User-centric component testing	DOM queries, user events
jsdom	Browser environment simulation	Node.js-based DOM

Test Coverage Areas

Area	What's Tested
Engines	Weight validation, scoring, ranking, cost estimation, SHAP additivity
API	All REST endpoints, error codes, response envelopes
Deployment	Dockerfile existence, render.yaml validity, static file serving
Observability	ETag generation, timing headers, metrics counters
Data Integrity	Coimbatore/Chennai separation, seed completeness

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10. Complete Version Matrix

Component	Package	Version	Category
React	`react`	19.2.8	Frontend Core
React DOM	`react-dom`	19.2.8	Frontend Core
Vite	`vite`	8.2.2	Frontend Build
Tailwind CSS	`tailwindcss`	4.3.3	Frontend Styling

Component	Package	Version	Category
React Router DOM	`react-router-dom`	7.18.3	Frontend Routing
Axios	`axios`	1.20.0	Frontend HTTP
Leaflet	`leaflet`	1.9.4	Frontend Map
React-Leaflet	`react-leaflet`	5.0.0	Frontend Map
Recharts	`recharts`	3.10.1	Frontend Charts
Three.js	`three`	0.185.1	Frontend 3D
Lucide React	`lucide-react`	1.42.0	Frontend Icons
jsPDF	`jspdf`	4.2.1	Frontend PDF
jsPDF AutoTable	`jspdf-autotable`	5.0.8	Frontend PDF
Vitest	`vitest`	5.0.0	Frontend Testing
oxlint	`oxlint`	1.79.0	Frontend Linting
FastAPI	`fastapi`	0.115.6	Backend Core
Uvicorn	`uvicorn`	0.34.0	Backend Server
Pydantic	`pydantic`	2.10.4	Backend Validation
Pydantic Settings	`pydantic-settings`	2.7.0	Backend Config
SQLAlchemy	`sqlalchemy`	2.0.36	Backend ORM
Alembic	`alembic`	1.14.0	Backend Migrations
psycopg	`psycopg`	3.2.3	Backend DB Driver
httpx	`httpx`	0.28.1	Backend HTTP
python-dotenv	`python-dotenv`	1.0.1	Backend Config
Pandas	`pandas`	3.0.5	Data Processing
NumPy	`numpy`	2.4.6	Data Processing
scikit-learn	`scikit-learn`	1.9.0	Machine Learning
XGBoost	`xgboost`	3.2.0	Machine Learning
SHAP	`shap`	0.51.0	Explainability
GeoPandas	`geopandas`	1.1.4	Geospatial
Shapely	`shapely`	2.1.2	Geospatial

Component	Package	Version	Category
ReportLab	`reportlab`	4.2.5	PDF Generation
paho-mqtt	`paho-mqtt`	2.1.0	IoT
PostgreSQL	—	16	Database
PostGIS	—	3.4	Database Extension
pgvector	—	latest	Database Extension
Docker	—	latest	Infrastructure
nginx	—	latest	Infrastructure
Mosquitto	—	2	Infrastructure

Total Dependencies: 19 frontend + 15 backend + 4 infrastructure services

All open source or free tier — no proprietary dependencies.

[END OF NIRMAN AI TECHNICAL STACK SPECIFICATION]